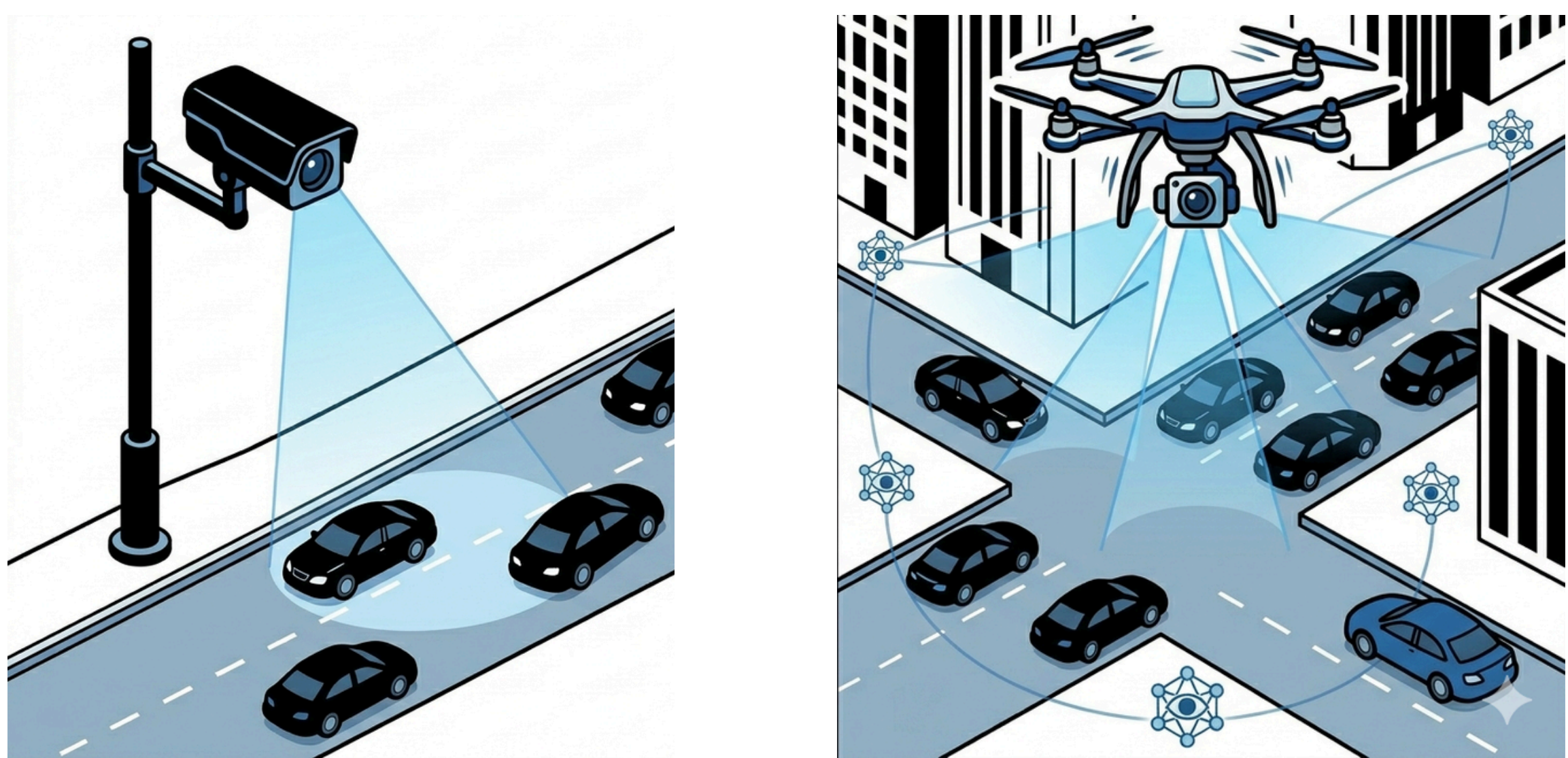


Introduction

This project proposes a Smart Drone Traffic Violation Detection System to overcome the limitations of traditional fixed surveillance systems, such as their restricted coverage and inability to effectively identify dynamic traffic offenses.



Methodology

An edge-based pipeline processes live drone footage to detect traffic violations in real time.

A drone equipped with a Raspberry Pi and camera captures real-time aerial video streams.

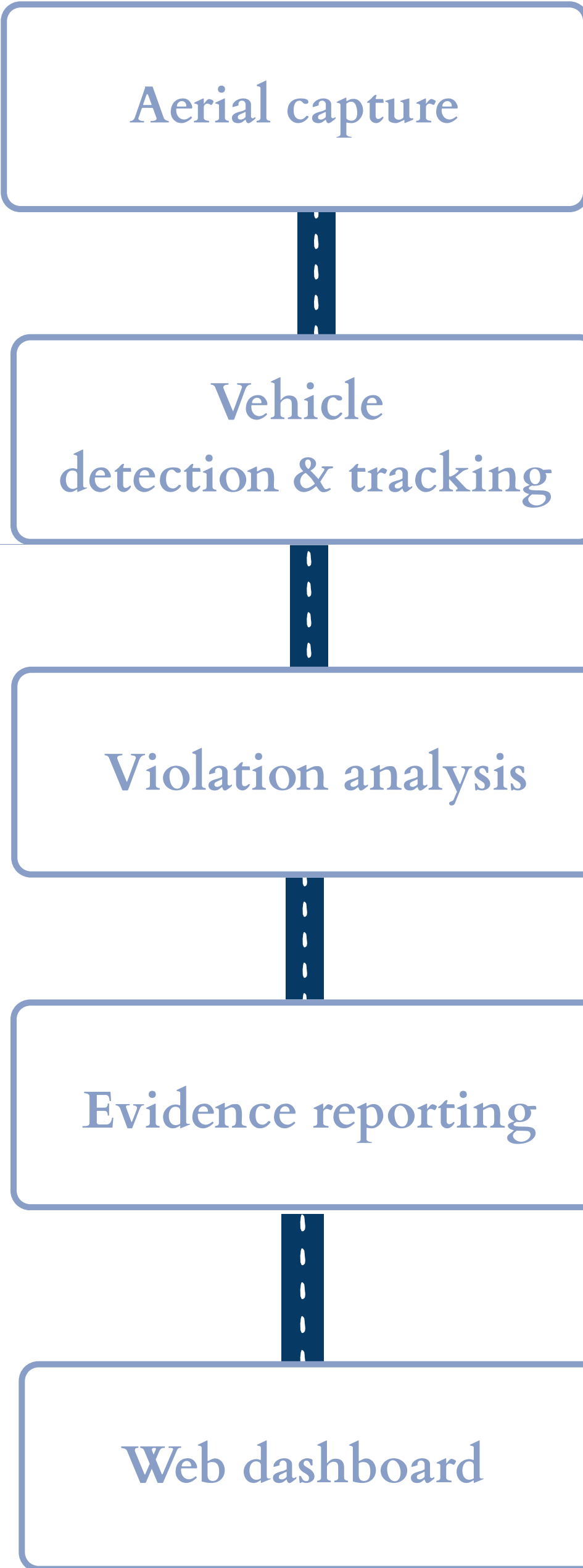
Vehicle detection is performed using YOLO26n, followed by multi-object tracking across consecutive frames using ByteTrack in combination with a Kalman filter

The system detects three types of violations:

- Wrong-way driving, based on trajectory and road orientation
- Dual-lane crossing, based on overlap with lane segmentation
- Missing license plates, using plate detection and OCR

For each violation, the system extracts the license plate, GPS location, timestamp, and snapshot, and stores them in Cloudflare R2 (image storage) and D1 (structured database).

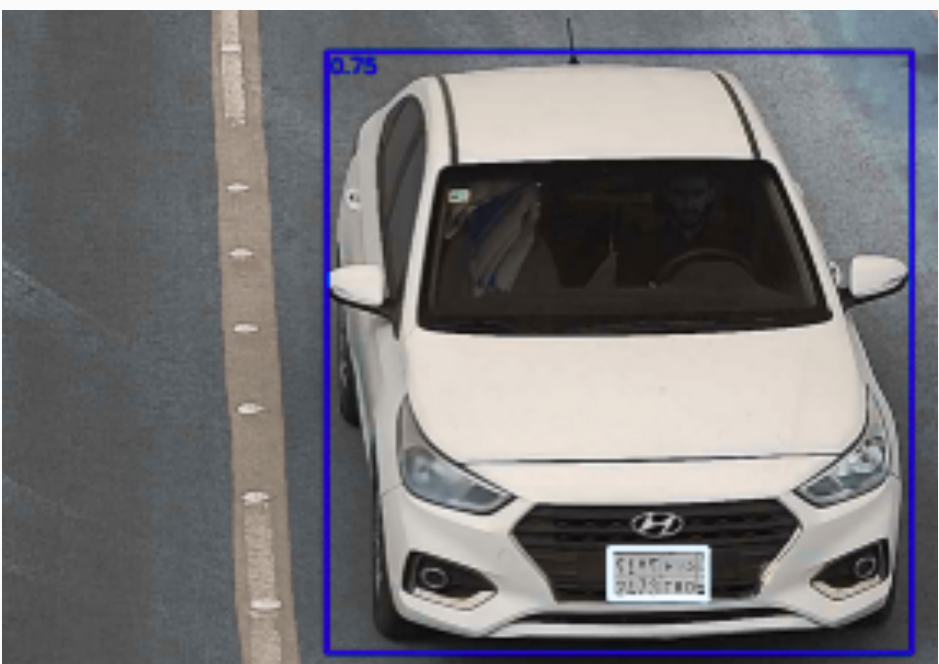
Finally, detected violations are streamed and visualized in real time through a web-based dashboard powered by FastAPI, utilizing HLS video streaming for continuous monitoring.



Experiments & Results

Detection

Simultaneous detection of vehicles, lane boundaries, and license plates using AI models



Detected Violations



Vehicle with missing plate



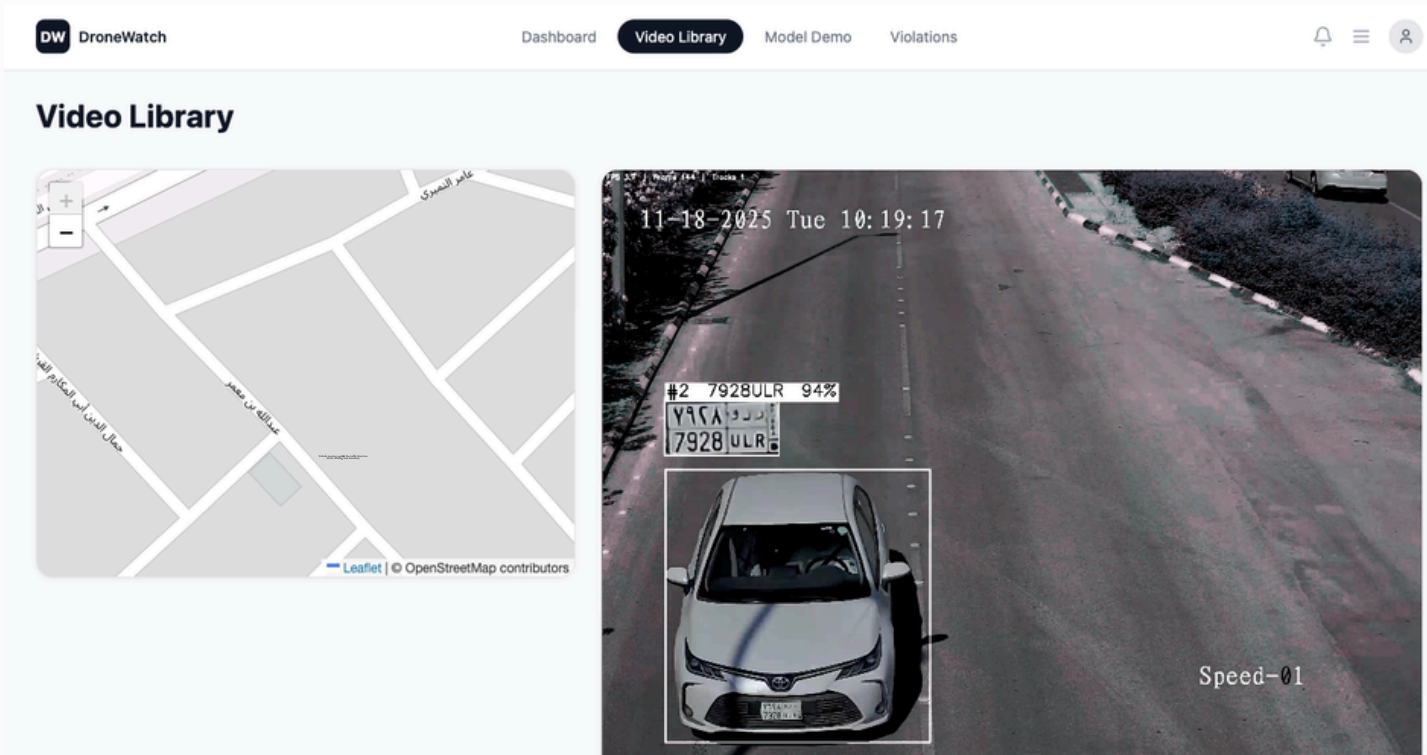
Vehicle crossing lane boundary



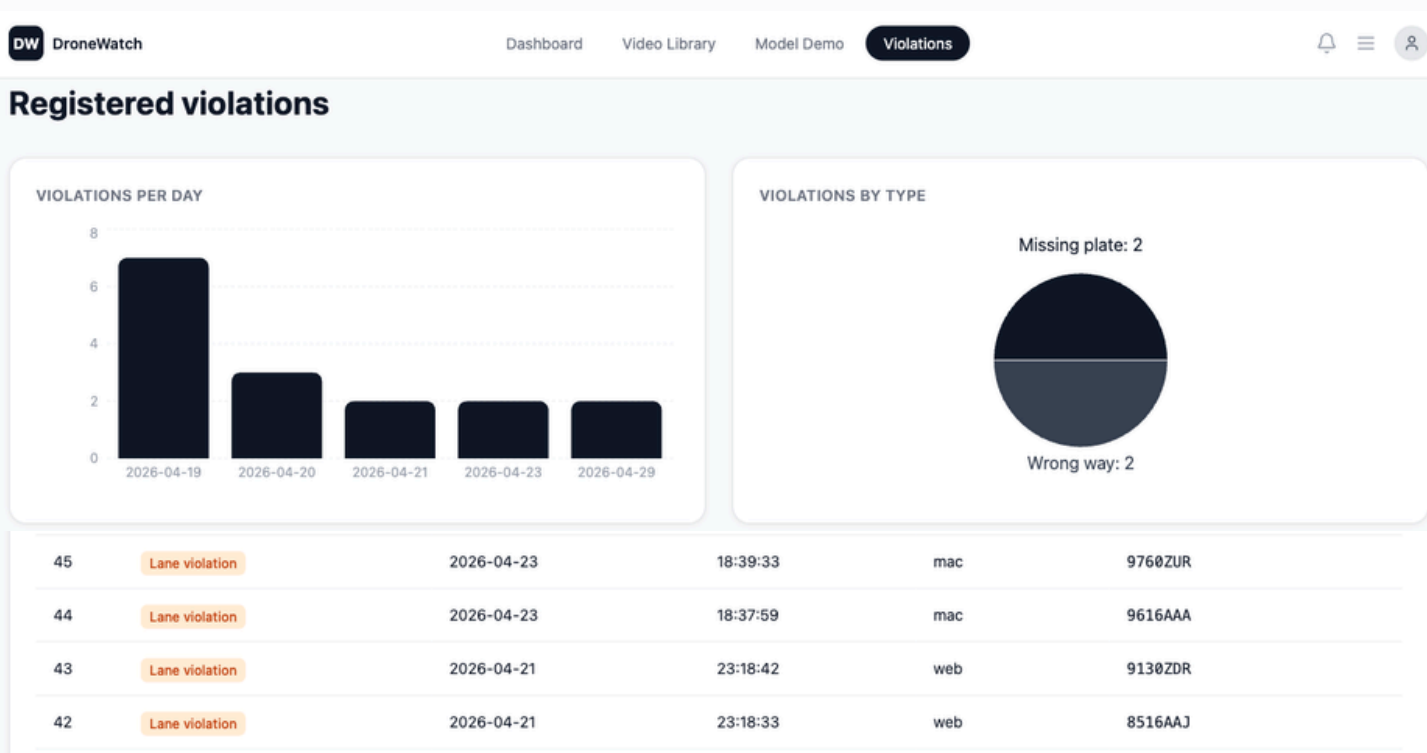
Vehicle moving against the traffic direction

System Interface

- Live stream viewer, detection insights, and interactive demo interface.



- Detected violations are stored with timestamps, source, and plate number



Conclusion

Innovative Framework

- Developed an autonomous UAV-based system integrating Pixhawk and Raspberry Pi for dynamic traffic surveillance.
- Pioneered a Hybrid Edge-Cloud Architecture that balances real-time on-board processing with high-performance cloud storage.



Technical Excellence

- Implemented a multi-stage AI pipeline using YOLO26n (Detection), ByteTrack (Tracking), and FastPlateOCR (Recognition).
- Achieved industry-leading accuracy: 96.4% Precision in plate detection and 95.1% Character Accuracy, ensuring reliable enforcement.



Strategic Impact & Vision 2030

- Successfully automated detection of complex violations: Wrong-Way Driving, Lane Infractions, and Missing Plates
- Delivers a scalable, cost-effective solution for smarter urban management, directly aligned with Saudi Vision 2030 for road safety.

